



Kickstarter Campaign Analysis

A Data Essay by Bryan Beus

Problem Identification

- Small-business owners sometimes have valuable ideas that require funding in order to become a reality
- Crowdsourcing is when an entrepreneur turns to the open internet for support for their ideas



Problem Identification

- A successful crowdsourcing campaign can help the entrepreneur, but a failed campaign can lead to a loss of time, resources, and social investment



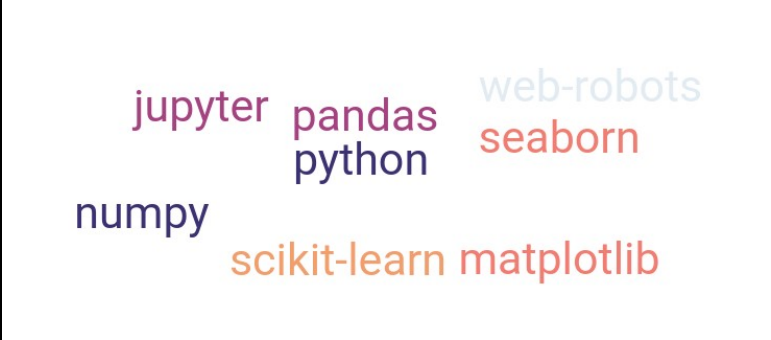
Project Context

- The data-driven essay assists crowdsourcing entrepreneurs through:
 - analytical techniques
 - an open-source dataset
 - predictive modeling



Solution Scope

- The scope of this essay includes tools such as:
 - Web Robots archive
 - Python
 - Jupyter
 - Numpy
 - Matplotlib
 - Scikit-Learn
 - Seaborn



A word cloud containing the names of various data science and machine learning tools. The words are arranged in a non-uniform, overlapping manner. The colors of the words are: jupyter (purple), pandas (dark blue), web-robots (light blue), python (dark blue), seaborn (red), numpy (dark blue), scikit-learn (orange), and matplotlib (red).

jupyter pandas web-robots
python seaborn
numpy scikit-learn matplotlib

Constraints

- Based on publicly available datasets:
 - Incomplete web-scrapings
 - Selection bias
- Only general categorization
 - Small-business owners may find that their project niche is difficult to relate

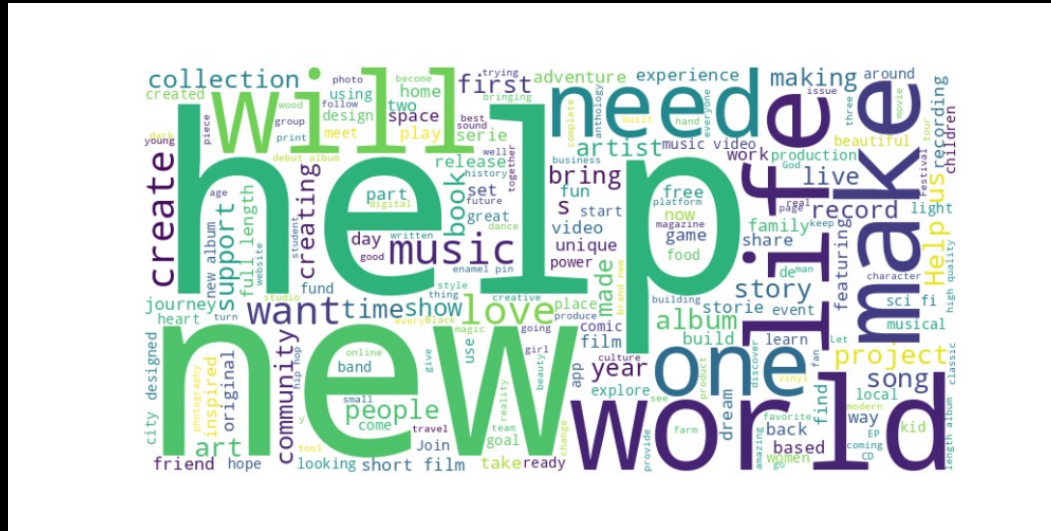


Data Wrangling

- **Kickstarter dataset**
 - Import to Jupyter
 - Convert json to pandas dataframe
 - Remove duplicates, null values
 - Drop unnecessary columns
 - Export as parquet file



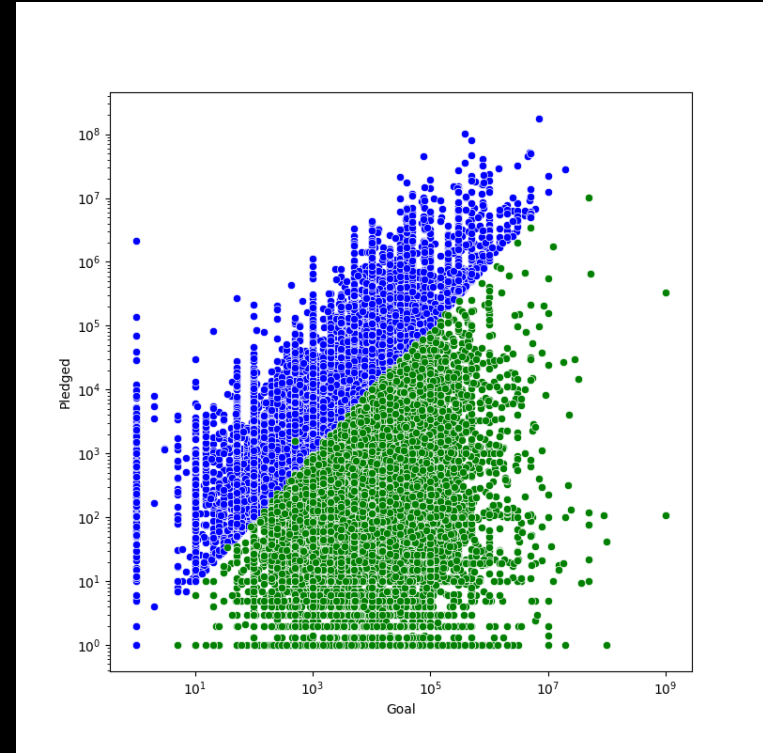
Exploratory Data Analysis



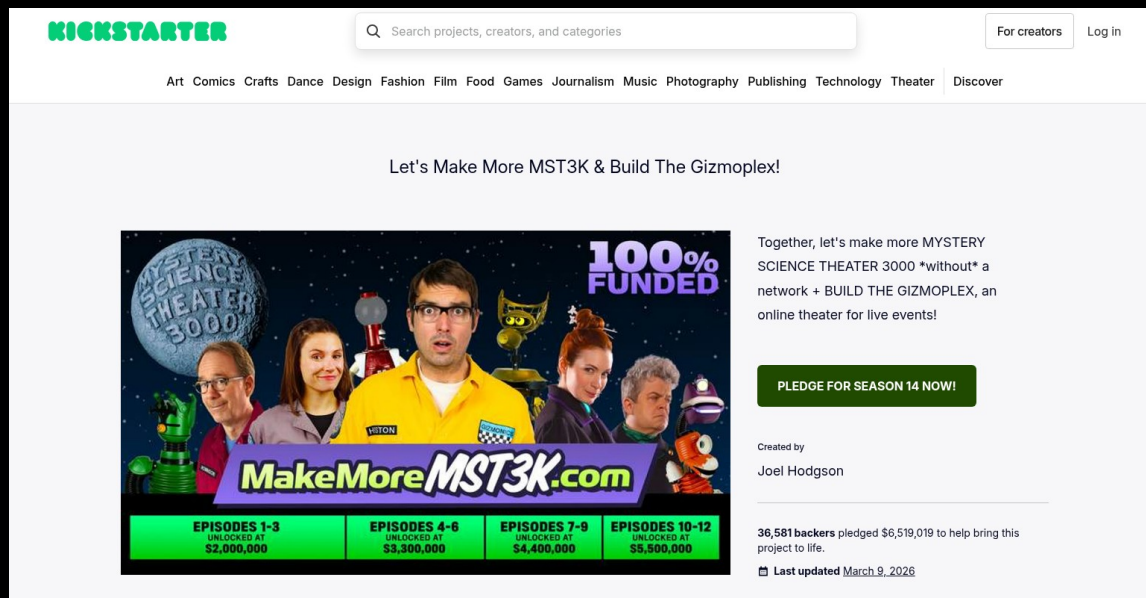
Word cloud based on common terms from Kickstarter project descriptions

Exploratory Data Analysis

- Goal vs Outcome
 - Successful campaigns in blue
 - Failed campaigns in green
- Note the some projects that had a goal of \$1 and received millions, as well as vice versa



Exploratory Data Analysis

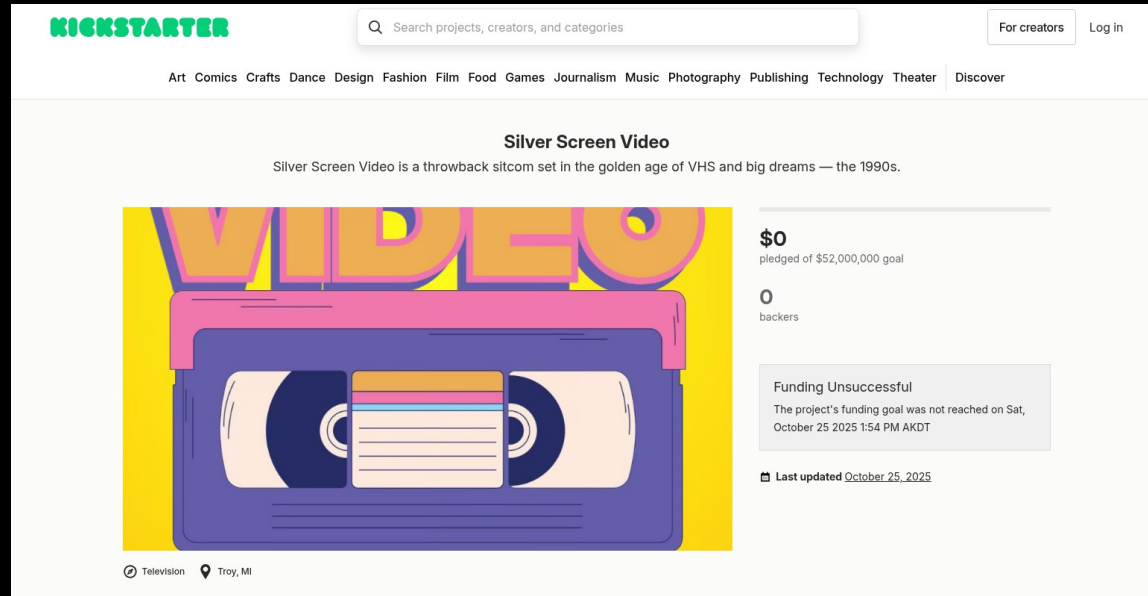


A notable project

Mystery Science Theater 3000

Goal: \$2M USD, Pledge: \$6.5M USD

Exploratory Data Analysis



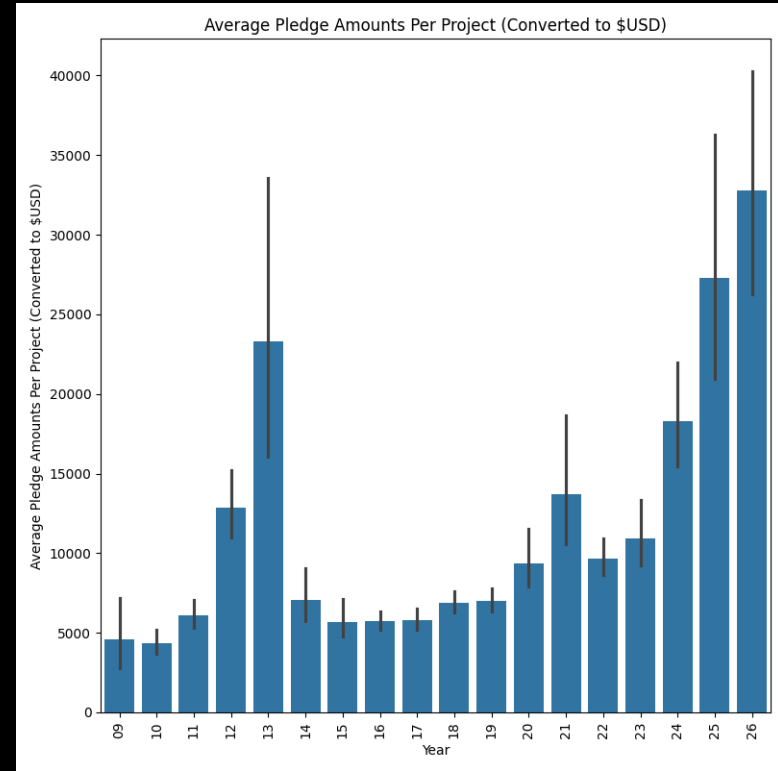
A notable project

Silver Screen Video

Goal: \$52M USD, Pledge: \$0 USD

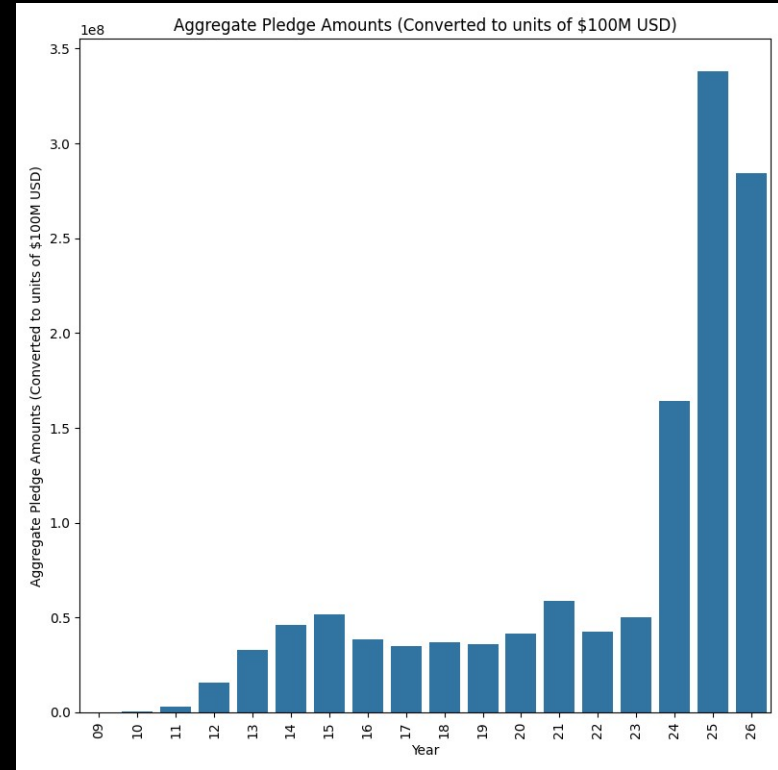
Exploratory Data Analysis

- Average pledged amounts per project are increasing
- (Note that this analysis could be subject to selection bias)



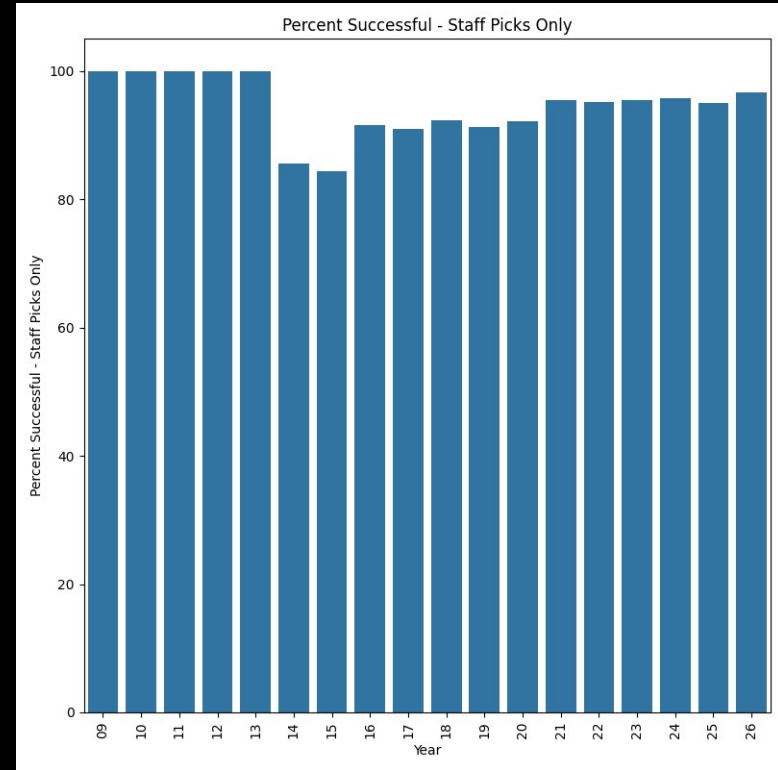
Exploratory Data Analysis

- Total pledged amounts are also increasing
- (Note possible selection bias)



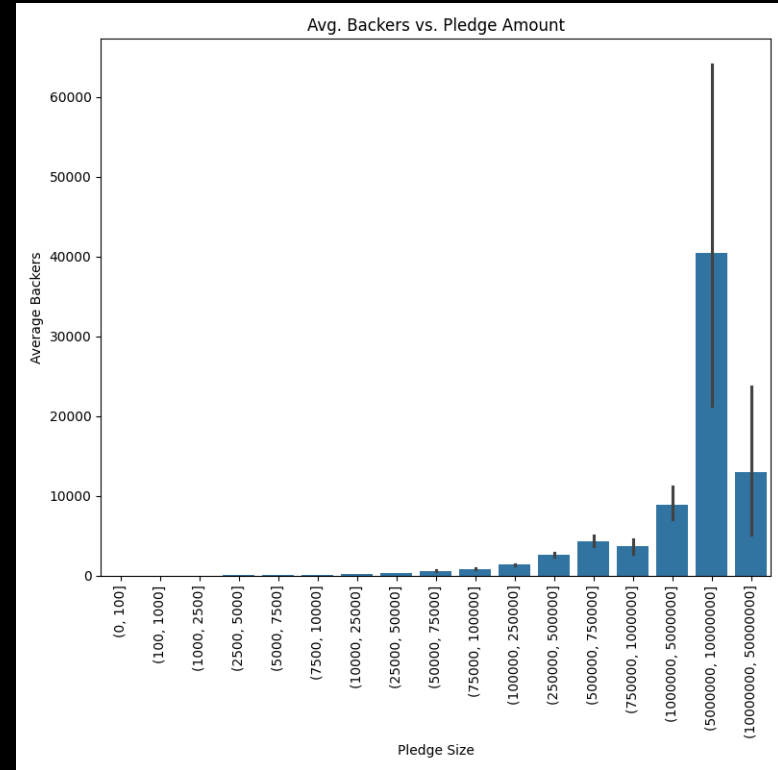
Exploratory Data Analysis

- Kickstarter projects that earn a “Staff Pick” highlight have an 80% chance or more of success



Exploratory Data Analysis

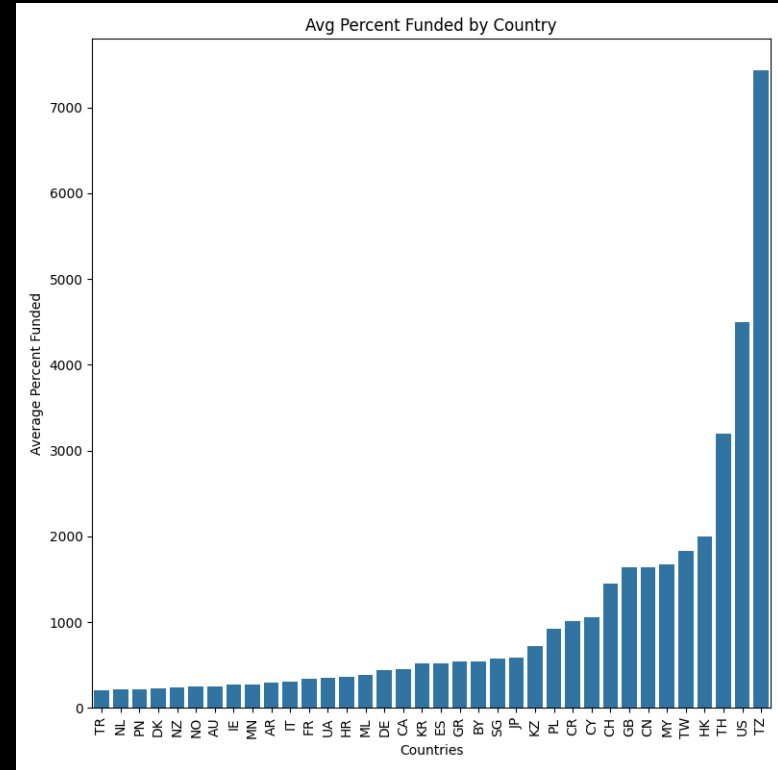
- Note that a project can earn as much as \$100K without requiring more than 1000 backers
- Even 350 backers can raise \$25K
- Seven-figure sums, however, require tens of thousands of backers



Exploratory Data Analysis

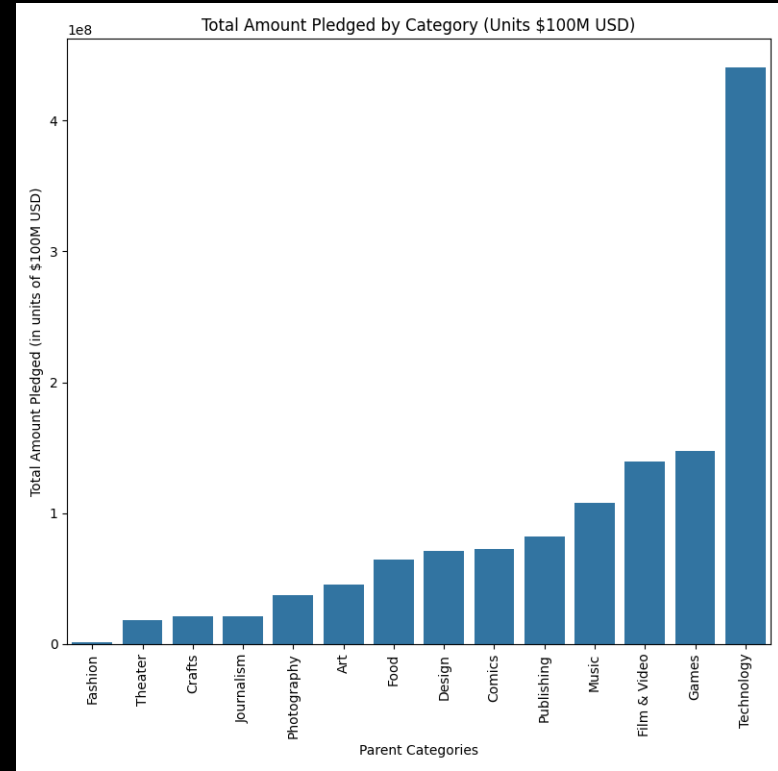
- Among successful projects, Tanzania is surprisingly the highest overachieving country, with an average of over 7000% project funding

This is likely due to a limited number of campaigns and skewed data



Exploratory Data Analysis

- The most lucrative category by far is Technology, followed by Games, Film & Video, and Music

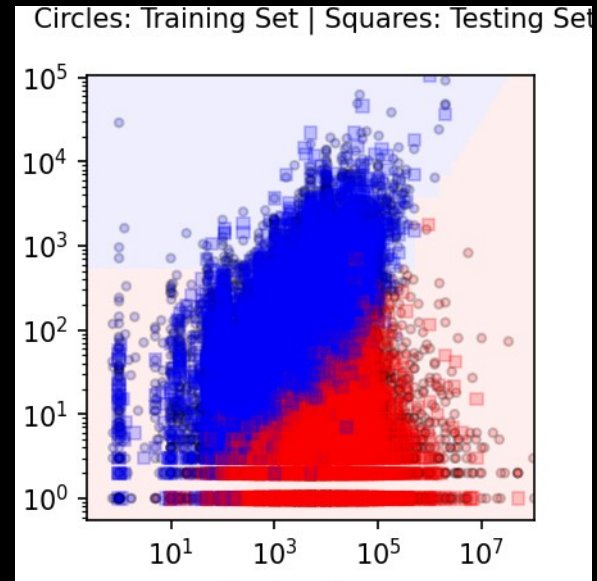


Exploratory Data Analysis

- Actionable Insights
 - Research how many committed donors beforehand
 - If only a few, keep goal at \$100 or below
 - Considerable sums of \$25000 < require 350+ backers

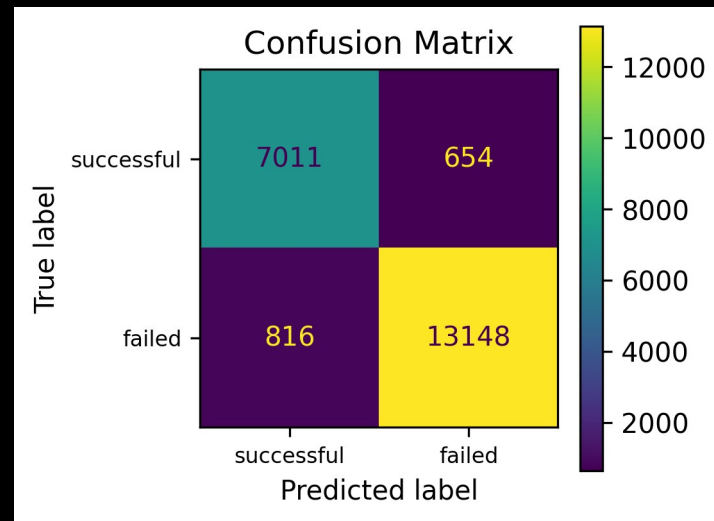
Predictive Modeling

- Logistic Regression Classification Decision Boundary Graph:
 - Successful campaigns in red
 - Failed campaigns in blue
 - x-axis: goal
 - Y-axis: pledge
- Note that successful campaigns are usually less ambitious



Predictive Modeling

- Logistic Regression Confusion Matrix
 - Overall accuracy: 0.92
- Logistic Regression model can likely provide a reasonably accurate prediction
- Notable features include:
 - Goal amount, expected backers, currency, location, category



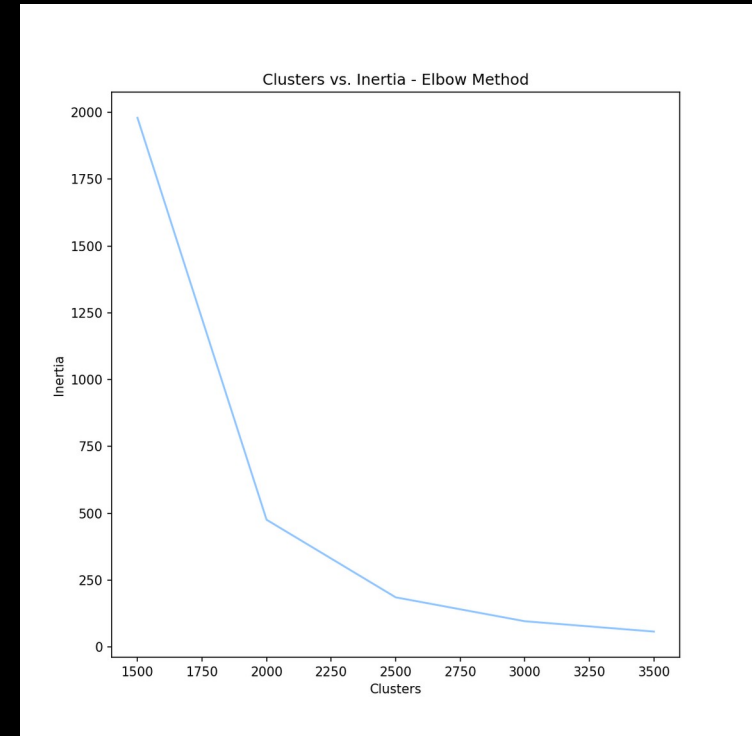
Predictive Modeling

K-Means Clustering

Total data rows: 86K

Inertia Elbow method:

Ideal number of clusters
is likely 2250



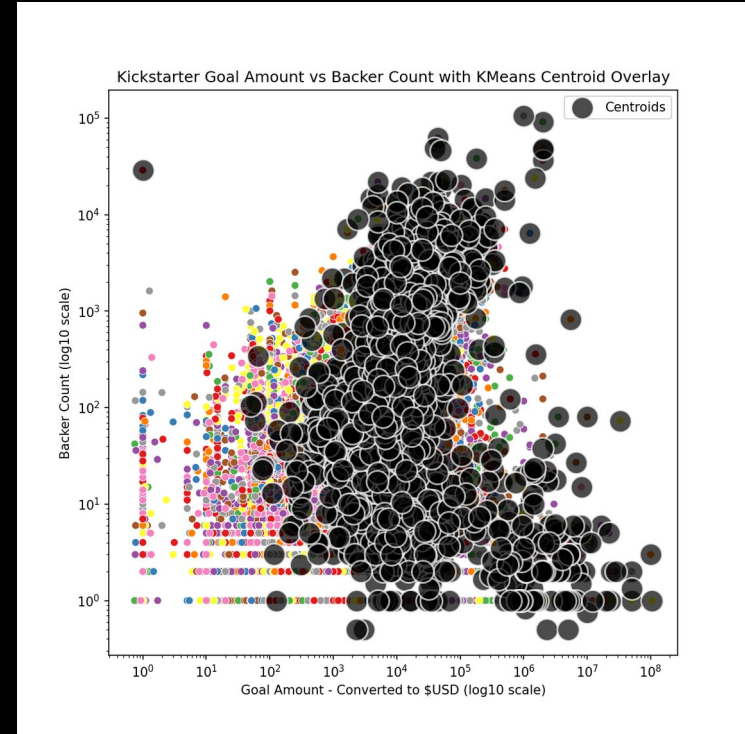
Predictive Modeling

Successful K-Means Cluster

Use case:

A Kickstarter candidate can put their parameters into the model and find similar projects in the same predicted cluster

This provides a useful tool for research and development



Conclusion

Recommendations and Insights:

- Keep project simple and low cost
- Research your number of committed supporters beforehand
- Limit your goal to a number that can be achieved
- Use K-Means Clustering to find similar projects and research their working models

To see a more complete version, visit:

- <https://github.com/blue-santa/kickstarter-campaign-analysis>
- <https://bluesanta.io>

Image Credits

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